

Lauren Elizabeth Gayle

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PROFESSIONAL SUMMARY

Computer Engineering student pursuing concurrent BS and MS degrees at Washington University in St. Louis, with hands-on research experience in autonomous systems and robotics at Carnegie Mellon and Oregon State. Proficient in C, Python, Java, and hardware description languages including VHDL and System Verilog, with coursework spanning embedded systems (RISC-V, AVR, MIPS), operating systems, parallel computing, and digital logic design.

EDUCATION

Washington University in St. Louis

BS in Computer Engineering. GPA: 3.7.

Expected May 2027

Relevant Coursework: Intro to Digital Logic (RISC-V, System Verilog), Intro to Computer Engineering (AVR), Elements of Computing Systems (VHDL), Operating Systems Organization (C), Signals and Systems, Computer Security, Technical Writing

Washington University in St. Louis

MS in Computer Engineering.

Expected May 2028

Rhodes College, Memphis, TN

BS in Computer Science. GPA: 3.8.

Degree Conferred May 2027

Relevant Coursework: Intro to Computer Organization and Architecture (MIPS), Intro to Computer Systems (C), Intro to Parallel Computing (OpenSHMEM, OpenMP, MPI, CUDA), Intro to Computer Networks (C), Data Structures and Algorithms, Programming Languages (Prolog, SML), Theory of Computation, Object Oriented Programming (Java)

EXPERIENCE

Teaching Assistant

January 2026 – Present

Washington University in St. Louis

St. Louis, MO

- Support students during in-class studios and office hours for the introductory Computer Engineering (Arduino C, Java, and AVR) and Digital Logic (RISC-V and System Verilog) courses.
- Grade assignments and provide individualized written feedback to reinforce student understanding.

Digitalization Intern

June 2026 – August 2026

BASF

Freeport, TX

- Migrated an MS Access database to an Excel-based platform to improve usability for operations teams.
- Developed Microsoft PowerApps applications to streamline the checkout of inspection equipment, site deficiency tracking, and construction requests.
- Created a Power BI dashboard to visualize ROI on purchased inspection equipment based on their average rental prices.

Digitalization Intern

June 2025 – August 2025

BASF

Freeport, TX

- Maintained and debugged an MS Access database used to log project information for design engineers.
- Updated UI and backend functionality of Microsoft PowerApps Request applications.
- Developed a paginated Power BI report from SharePoint lists to visualize KPI information.

Robotics Institute Intern

June 2024 – May 2025

*Carnegie Mellon University, Robotics Institute**Pittsburgh, PA*

- Investigated robustness testing for autonomous vehicle software, identifying rare and off-nominal events in system logs including speeding incidents and scene misinterpretations.
- Developed a Python-based runtime monitor to detect and flag unsafe driving behaviors in self-driving car logs, targeting edge cases that caused hazardous system states.

Undergraduate Researcher

June 2023 – August 2023

*Oregon State University**Corvallis, OR*

- Authored a literature survey citing 30 papers on adaptive behavior trees to support a PhD student's research.
- Implemented a library of 8 primitive motion actions in Python and ROS for the GoBot, a TurtleBot-based assistive robot designed for early childhood motor interventions.

Research Assistant

January 2023 – May 2023

*Rhodes College**Memphis, TN*

- Tuned a Python-based computer vision software used in live demonstrations for explainable robotics research targeted at policy audiences and lawmakers.

Computer Science Grader

January 2025 – May 2025

*Rhodes College**Memphis, TN*

- Evaluated python project submissions for introductory computer science courses and delivered individualized written feedback.

Computer Science Peer Tutor

August 2023 – May 2025

*Rhodes College**Memphis, TN*

- Tutored peers across three introductory Computer Science courses: Intro to Programming (Python), Intro to Object Oriented Programming (Java), and Data Structures and Algorithms (Java).
- Adapting instructional approach to each student's learning style and concept.

ACTIVITIES & LEADERSHIP

WashU Robotics, Nimble | Software Team Member

- Contribute to the robotic arm project, Nimble, by researching the application of vision-language-action models (VLAs) for robotic controls.

Rhodes College Gender and Sexuality Alliance | PR Chair

- Designed promotional materials and social media content for events; coordinated outreach to performers and venues.

Rhodes College Women in STEM Club | Secretary

- Recorded meeting minutes for leadership board sessions and distributed monthly communications to club members.

Rhodes College Fencing Club | Social Media Coordinator

- Created social media content to promote club meetings and tournament participation.

SKILLS

Languages: C, Python, Java, Prolog, SML**Hardware Description Languages:** VHDL, System Verilog**Embedded Architectures:** RISC-V, AVR, MIPS**Parallel Computing:** OpenMP, MPI, CUDA, OpenSHMEM**Tools and Platforms:** ROS, Microsoft PowerApps, Power BI, SharePoint, MS Access